Bounds for 3-AP-Free Sets (Kelley–Meka Bounds)

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   - Motivating Question
   - Historical Background
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Overview: Motivating Question

Problem

How large a subset $A \subseteq S_N := \{1, 2 \ldots, N\}$ can be such that it contains no non-trivial three-term[a] arithmetic progression

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- The problem statement is merely simple, but it has become one of the central questions in additive combinatorics.
- The tools and techniques required to tackle the problem is the reason of the significance of this problem.

Overview: Historical Background

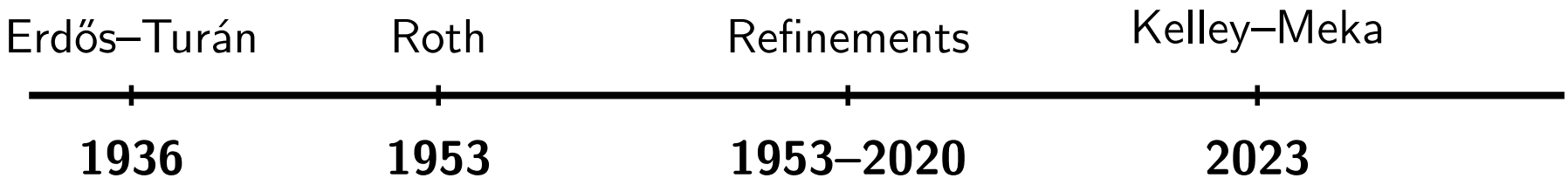


Figure: Chronology of 3-AP-free set bounds.

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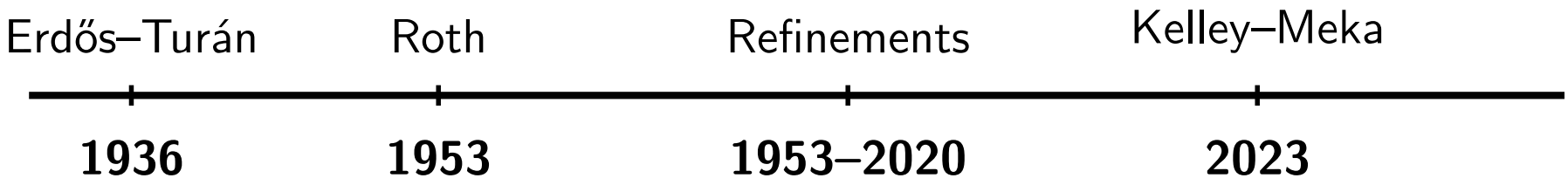


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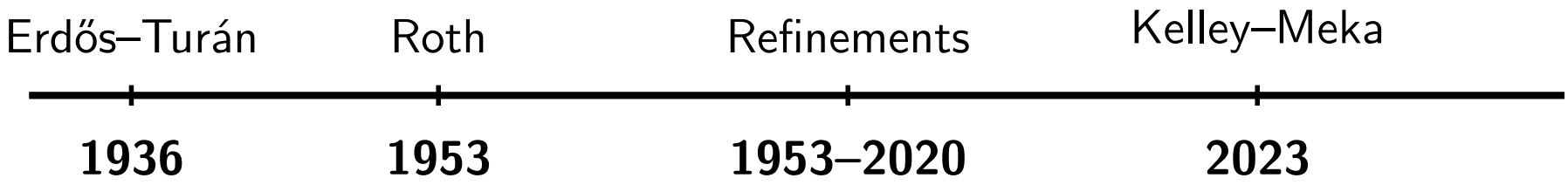


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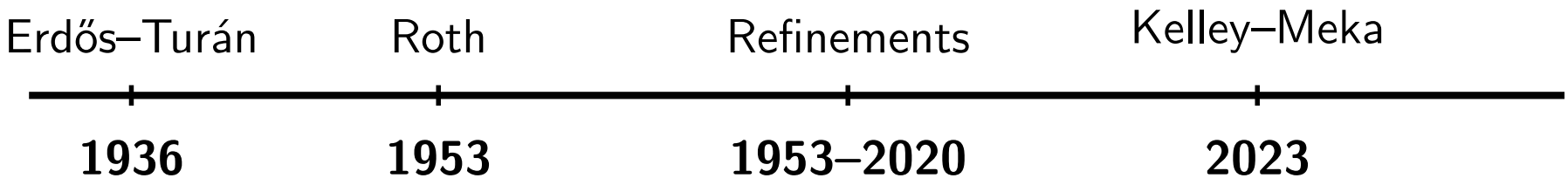


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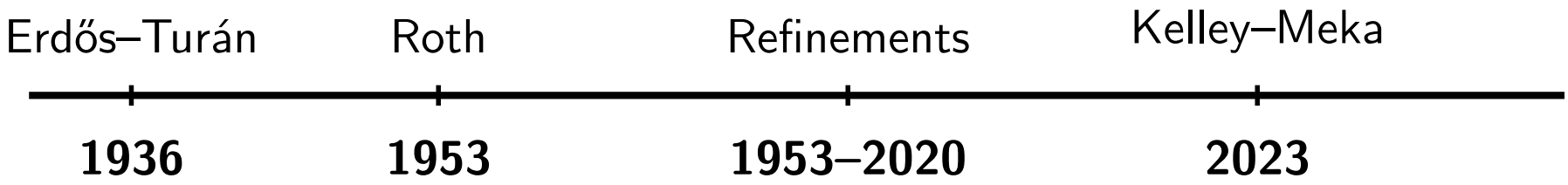


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- **2023:** Breakthrough by Kelley–Meka.

Overview: Main Result

Theorem 1. (Kelley-Meka)

If $A \subseteq S_N$ contains no non-trivial three-term arithmetic progression, then

$$|A| \leq \frac{N}{\exp(c(\log N)^{1/12})}$$

for some absolute constant $c > 0$.

Overview: Strategy

Work in Finite field $\mathbb{F}_q^n$ (Theorem 2) $\rightsquigarrow$ Integer case S_N (Theorem 1)

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Theorem 2 (Kelley-Meka)

If q is an odd prime and $A \subseteq \mathbb{F}_q^n$ has no non-trivial three-term progressions, then

$$|A| \leq q^{n - cn^{1/9}}$$

for some constant $c > 0$.

Notational Conventions (1)

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- **(Relative) Inner product and norm L^p.**
  Suppose $\mu : G \to \mathbb{R}_{\geq 0}$ s.t $\|\mu\|_1 = 1$. For functions $f, g : G \to \mathbb{C}$, define

$$\langle f, g\rangle_\mu := \mathop{\mathbb{E}}_{x\in G} \mu(x) f(x)\overline{g(x)},$$

  and

$$\|f\|_{p(\mu)} := \left(\mathop{\mathbb{E}}_{x\in G} \mu(x)|f(x)|^p\right)^{1/p}, \qquad 1 \leq p < \infty.$$

  In particular, when $\mu \equiv 1$, we write $\langle f, g\rangle$ and $\|f\|_p$.

Notational Conventions (2)

- **Convolution and Difference Convolution.** The convolution and difference convolution of f and g are defined by

$$f * g(x) := \mathop{\mathbb{E}}_{y \in G} f(y)\, g(x-y), \text{ and } f \circ g(x) := \mathop{\mathbb{E}}_{y \in G} f(x+y)\, g(y),$$

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- We define $\mathcal{L}(\alpha) := \log(2/\alpha)$.

Finite Field $\mathbb{F}_q^n$ Case: Starting Point

- Let $A, C \subseteq F_q^n$ be sets with density α, γ, respectively. We deploy $\langle \mu_A * \mu_A, \mu_C \rangle$ to count the expected number of solution to $a_1 + a_2 = c$, where $a_1, a_2 \in A$, $c \in C$.

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- This is because

$$\begin{aligned}
\langle \mu_A * \mu_A, \mu_C \rangle &= \alpha^{-2}\gamma^{-1}\mathbb{E}_{x\in\mathbb{F}_q^n}\mathbb{E}_{y\in\mathbb{F}_q^n} 1_A(y)\, 1_A(x-y)\, 1_C(x) \\
&= \alpha^{-2}\gamma^{-1}|\mathbb{F}_q^n|^{-2} \sum_{c\in C} \#\{a_1, a_2 \in A : a_1 + a_2 = c\} \\
&= \alpha^{-2}\gamma^{-1}|\mathbb{F}_q^n|^{-2} \#\{(a_1, a_2, c) \in A \times A \times C : a_1 + a_2 = c
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- Therefore, if A and C are random set, the number of solution to the equation (after normalization) is

$$\langle \mu_A * \mu_A, \mu_C \rangle \approx 1.$$

Finite Field $\mathbb{F}_q^n$ Case: Starting Point (2)

- We assume that $A \subseteq \mathbb{F}_q^n$ contains no non-trivial three-term arithmetic progressions. Hence,

$$\#(\text{3-AP in } A) = |A|.$$

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- We now begin with a constant discrepancy:

$$\langle \mu_A * \mu_A, \mu_C \rangle \leq 1 - \delta,$$

for some constant $\delta > 0$.

Kelley-Meka Strategy: 5 steps

1. Hőlder Lifting

If $\langle \mu_A * \mu_A, \mu_C \rangle \leq 1/2$, then $||\mu_A \circ \mu_A - 1||_p \geq 1/4$ for some $p \ll \mathcal{L}(\gamma)$.

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2. Unbalancing

For any $f : G \to \mathbb{R}$ such that $\widehat{f} \geq 0$, if $\|f\|_p \geq 1/4$, then $\|f + 1\|_{p'} \geq 1 + 1/8$ for some $p' \ll p$.
In particular, if $\|\mu_A \circ \mu_A - 1\|_p \geq 1/4$, then $\|\mu_A \circ \mu_A\|_{p'} \geq 1 + 1/8$.

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3. Dependent Random Choice

If $\|\mu_A \circ \mu_A\|_p \geq 1 + 1/8$, then there are $A_1, A_2 \subseteq A$ of density at least $\alpha^{O(p)}$ such that

$$\langle \mu_{A_1} \circ \mu_{A_2}, 1_S \rangle \geq 1 - 1/32,$$

where $S = \{x \in G : \mu_A \circ \mu_A(x) > 1 + 1/16\}$.

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4. Almost Periodicity

For any sets $A_1, A_2, S \subseteq \mathbb{F}_q^n$, if A_1, A_2 have density at least α, then there is a subspace V with codimension $O(\mathcal{L}(\alpha)^4)$ such that

$$\left|\langle \mu_V * \mu_{A_1} \circ \mu_{A_2}, 1_S \rangle - \langle \mu_{A_1} \circ \mu_{A_2}, 1_S \rangle\right| \leq 1/100.$$

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5. Density Increment

If $\langle \mu_A * \mu_A, \mu_C \rangle \leq 1/2$, then there is an affine subspace V of codimension $O(\mathcal{L}(\alpha)^4 \mathcal{L}(\gamma)^4)$ on which A has density at least $\left(1 + \frac{1}{100}\right)\alpha$.

Proof of Theorem 2 (Assuming Step 5)

Proof. Choose $C = 2 \cdot A$. Then Step 5 states:
If $\langle \mu_A * \mu_A, \mu_{2A} \rangle \leq \frac{1}{2}$, then there exists an affine subspace V of codimension $O(\mathcal{L}(\alpha)^8)$ on which a translate of A has density at least $\left(1 + \frac{1}{100}\right)\alpha$.

- Starting from $A_0 := A \subseteq \mathbb{F}_q^n$, $V_0 := F_q^n$, $A_i \subseteq V_i$ for all i; we iterate:

$$A_{i+1} = (A_i - x_i) \cap V_{i+1}, \quad V_{i+1} \subseteq V_i.$$

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- At this point we must have

$$\langle \mu_{A_m} * \mu_{A_m}, \mu_{2A_m} \rangle > \frac{1}{2}.$$

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- Therefore, $\alpha \leq q^{-cn^{1/9}}$ and hence $|A| \leq q^{n-cn^{1/9}}$. □

Moving from $\mathbb{F}_q^n$ to S_N

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- In $\mathbb{F}_q^n$, we have a *subspace*. However, in S_N, such structure is no longer available.
- To overcome this, we introduce *Bohr sets*, which will serve as a replacement for subspaces in this setting.

From Subspace to Bohr set (1)

Theorem 4 (Kelley–Meka)

Let $\varepsilon > 0$. There is a constant $C = C(\varepsilon) > 0$ such that the following holds. Let $G = \mathbb{F}_q^n$ for some prime q and $n \geq 1$. Let $p \geq 1$. For any non-empty $A \subseteq G$ with $|A| = \alpha|G|$ there exists a subspace $V \leq G$ with

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2. $\|\mu_{A'} * \mu_{A'} - \mu_V\|_{p(\mu_V)} \leq \varepsilon \dfrac{|G|}{|V|}$.

From Subspace to Bohr set (2)

Definition 1 (Bohr set).

For a non-empty $\Gamma \subseteq \widehat{G}$ and $\nu \in [0, 2]$ we define the Bohr set $B = \mathrm{Bohr}_\nu(\Gamma)$ as

$$\mathrm{Bohr}_\nu(\Gamma) = \{x \in G : |1 - \gamma(x)| \leq \nu \text{ for all } \gamma \in \Gamma\}.$$

We call Γ the *frequency set* of B and ν the *width*, and define the *rank* of B to be the size of Γ, denoted by $\mathrm{rk}(B)$.

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Definition 2 (Regularity of Bohr set).

A Bohr set B of rank d is *regular* if for all $|\kappa| \leq \frac{1}{100d}$ we have

$$(1 - 100d|\kappa|)\,|B| \leq |B_{1+\kappa}| \leq (1 + 100d|\kappa|)\,|B|.$$

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Fact 3 (Rank vs Codimension)

A Bohr set of rank d behaves like a subspace of codimension d.

Theorem 4 $\rightsquigarrow$ Theorem 5

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For any finite abelian group G and any subset $A \subseteq G$ with $|A| = \alpha|G|$, there exists a regular Bohr set B with

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3. $||(\mu_{A'} - \mu_B) * (\mu_{A'} - \mu_B)||_{p(\mu_{k \cdot B'})} \leq \epsilon \dfrac{|G|}{|B|}$.

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$$|A| \leq N \exp(-c(\log N)^{1/12}).$$

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